

Perfect difference sets: i.o. lower bounds, exact small- N optima, and an extension obstruction

Notes on Erdős Problem #1194

Mehmet Ergezer

Wentworth Institute of Technology, Boston, MA

ergezerm@wit.edu ORCID: 0000-0001-6627-3667

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Abstract

A set $A \subset \mathbb{N}$ is a perfect difference set (PDS) if every $n \geq 1$ admits a unique representation $n = a_n - b_n$ with $a_n, b_n \in A$. Erdős asked how fast a_n/n must increase [1]. We give four contributions. First, we record full proofs of the infinitely-often lower bounds $a_n \gg n \log n$ (classical), $a_n \gg n^{2-o(1)}$ (GPT-5.4 Pro / Bloom, April 2026), and the logarithmic refinement $a_n \gg n^2/f(n)$ for any nondecreasing slowly varying f with $\sum 1/(nf(n)) < \infty$. Second, we compute exactly the value $M^*(N) := \min\{\max(A) : A \text{ Sidon}, [1, N] \subset A - A\}$ for $N \leq 45$ via branch-and-bound and CP-SAT, observing $M^*(N) \approx 2N$. Third, we exhibit an empirical *extension obstruction*: the optimum $A_{N_0}^*$ for the finite problem is a poor starting seed for an infinite PDS, with forced extensions growing like N^{4+} , worse than Lev’s greedy. We formulate a corresponding “for all large n ” conjecture and discuss why the partition-identity argument is sharp at the $n^{2-o(1)}$ ceiling. Finally, we report two systematic negative results: every block-structured seeding strategy we tested (split Mian–Chowla, arithmetic progressions of Sidon chunks) leaves the empirical $\Theta(N^3)$ greedy ceiling intact, and three structural attempts to improve the Sidon density bound specifically for PDS (Cauchy–Schwarz on the partition identity, Fejér-weighted Plancherel, higher L^p moments) each collapse to the generic Sidon bound. Code and data accompanying this note live at https://github.com/memoatwit/erdos_1194.

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1 Setup and notation

A set $A \subset \mathbb{N}$ is a *perfect difference set* (PDS) if every $n \geq 1$ admits a *unique* representation

$$n = a_n - b_n, \quad a_n, b_n \in A, \quad a_n > b_n.$$

The notation a_n is fixed: it is the larger element of the unique representing pair for n . It is *not* the n -th element of A . To avoid confusion we write the increasing enumeration of A as

$$A = \{x_1 < x_2 < x_3 < \cdots\}.$$

A PDS is in particular a Sidon set: all pairwise differences are distinct. We write $A_x := A \cap [1, x]$, $k(x) := |A_x|$, and $N(x) := \#\{n \geq 1 : a_n \leq x\}$. The running maximum is $M(N) := \max_{n \leq N} a_n = \max(\bigcup_{n \leq N} \{a_n, b_n\})$ when A is built greedily from $n = 1$ outward.

2 The PDS counting identity

Lemma 1 (PDS counting). *For every $x \geq 1$, $N(x) = \binom{k(x)}{2}$.*

Proof. Each $n \leq N(x)$ has $a_n \leq x$, hence both $a_n, b_n \in A_x$. The map $n \mapsto \{a_n, b_n\}$ is injective into unordered pairs of A_x , so $N(x) \leq \binom{k(x)}{2}$. Conversely, every unordered pair $\{a, b\} \subset A_x$, $a > b$, gives a difference $a - b \in [1, x - 1]$ which by uniqueness is the representation of some n with $a_n = a$, $b_n = b$, $a_n \leq x$. Hence $\binom{k(x)}{2} \leq N(x)$. $\square$

This identity will be used repeatedly. As a corollary, $\binom{k(x)}{2} \leq x$, recovering the standard Sidon density bound $k(x) \leq (1 + o(1))\sqrt{2x}$ as an inequality only — but PDS gives an *equality* $N(x) = \binom{k(x)}{2}$, not merely an inequality.

3 Infinitely-often lower bounds

3.1 The classical $n \log n$ bound

Theorem 1 (Erdős; classical). *For any infinite PDS A , $a_n \gg n \log n$ for infinitely many n .*

Proof. By Erdős's i.o. Sidon density bound (see e.g. Halberstam–Roth [2], Ch. II), there exist $x_j \rightarrow \infty$ with $k(x_j) \leq C\sqrt{x_j/\log x_j}$. By Lemma 1, $N(x_j) = \binom{k(x_j)}{2} \leq C^2 x_j / (2 \log x_j)$. Setting $n_j := N(x_j) + 1$, the element $a_{n_j} > x_j$ and $n_j \leq C^2 x_j / (2 \log x_j) + 1$, so

$$a_{n_j} > x_j \geq \frac{2 \log x_j}{C^2} (n_j - 1) \geq \frac{2(1 + o(1))}{C^2} n_j \log n_j. \quad \square$$

3.2 The partition identity and $n^{2-o(1)}$

The next theorem improves Theorem 1 to almost- n^2 via a completely different argument due to GPT-5.4 Pro and condensed by T. Bloom in the comment thread of erdosproblems.com/1194 (post 5745, April 23 2026). The argument does not use Sidon density at all; it uses the following partition identity together with a gap lower bound.

Lemma 2 (PDS partition). *For each $k \geq 2$ let $D_k := \{x_k - x_i : 1 \leq i < k\}$. Then $\{D_k\}_{k \geq 2}$ is a partition of $\mathbb{Z}_{>0}$. Consequently, for every $X \geq 1$,*

$$X = \sum_{k \geq 2} |D_k \cap [1, X]|.$$

Proof. Every $n \geq 1$ has a unique representation $n = a_n - b_n$. Pick k with $a_n = x_k$ and i with $b_n = x_i$; then $n = x_k - x_i \in D_k$. Uniqueness of the representation gives uniqueness of k . $\square$

Lemma 3 (Gap bound from $a_n \ll n^c$). *Suppose $a_n \leq C n^c$ for all $n \geq 1$ and some $c > 1$, $C > 0$. Then $x_k - x_{k-1} \geq (x_k/C)^{1/c}$ for every $k \geq 2$, and consequently $x_k \gg k^{c/(c-1)}$.*

Proof. Let $d_k := x_k - x_{k-1}$. The pair (x_k, x_{k-1}) realises difference d_k , so by PDS uniqueness $a_{d_k} = x_k$. Hence $x_k \leq C d_k^c$ and $d_k \geq (x_k/C)^{1/c}$. Telescoping, $x_k \geq \sum_{i=2}^k d_i \geq C^{-1/c} \sum_i x_i^{1/c}$, which is dominated by $x_k \gg k^{c/(c-1)}$ by an elementary comparison with the ODE $dy/dk = c_0 y^{1/c}$. $\square$

Theorem 2 (GPT-5.4 Pro, after Bloom). *For any infinite PDS A and any $c \in (1, 2)$, the inequality $a_n \leq C n^c$ for all n is impossible (for any C). Equivalently, $\limsup_{n \rightarrow \infty} a_n/n^c = \infty$ for every $c < 2$, i.e. $a_n \gg n^{2-o(1)}$ infinitely often.*

Proof. Suppose, for contradiction, that $a_n \leq C n^c$ for all n and some $c \in (1, 2)$. By Lemma 3, $x_k \gg k^{c/(c-1)}$.

Fix $X \geq 1$ and let $K = \lfloor X^{(c-1)/c} \rfloor$. By Lemma 2,

$$X = \underbrace{\sum_{2 \leq k \leq K} |D_k \cap [1, X]|}_{\text{(I)}} + \underbrace{\sum_{k > K} |D_k \cap [1, X]|}_{\text{(II)}}.$$

For (I), trivially $|D_k| \leq k - 1$, so $\text{(I)} \leq K^2/2 \ll X^{2-2/c}$.

For (II), $k > K$ gives $x_k > 2X$ for X large (absorb a constant into K). Any consecutive $x_{j-1} < x_j$ in $A \cap [x_k - X, x_k]$ satisfies $x_j - x_{j-1} \geq (x_j/C)^{1/c} \gg x_k^{1/c}$ by Lemma 3 and $x_j \geq x_k/2$. Hence $|D_k \cap [1, X]| \leq X/x_k^{1/c} \ll X \cdot k^{-1/(c-1)}$. Since $1/(c-1) > 1$, the sum $\sum_{k > K} k^{-1/(c-1)}$ converges with tail $\ll K^{(c-2)/(c-1)}$, giving $\text{(II)} \ll X \cdot K^{(c-2)/(c-1)} \ll X^{2-2/c}$.

Combining, $X \ll X^{2-2/c}$, i.e. $X^{2/c-1} \ll 1$ uniformly. Since $c < 2$, $2/c - 1 > 0$ and the left side $\rightarrow \infty$. Contradiction. $\square$

3.3 Logarithmic refinement

The boundary $c = 2$ in Theorem 2 is sharp; the same scheme under the slightly weaker hypothesis $a_n \leq Cn^2/f(n)$ pushes the bound to the convergence boundary of $\sum 1/(nf(n))$.

Theorem 3 (Logarithmic refinement). *Let $f: \mathbb{N} \rightarrow \mathbb{R}_{>0}$ be nondecreasing, slowly varying (i.e. $f(\lambda n)/f(n) \rightarrow 1$ for every fixed $\lambda > 0$), unbounded, with $\sum_{n \geq 2} \frac{1}{nf(n)} < \infty$. Then for any infinite PDS A ,*

$$a_n \gg \frac{n^2}{f(n)} \quad \text{for infinitely many } n.$$

In particular $a_n \gg n^2/(\log n)^{1+\varepsilon}$ i.o. for every $\varepsilon > 0$, and $a_n \gg (n/\log n)^2$ i.o.

Proof sketch. Replace $a_n \leq Cn^c$ by $a_n \leq Cn^2/f(n)$. The gap bound becomes $d_k \geq \sqrt{x_k f(d_k)/C}$, and for slowly varying f this iterates to $x_k \gg k^2 f(k)$. The anchor window bound in Step (II) becomes $|D_k \cap [1, X]| \leq X/(kf(k))$, and the partition sum split at $K = \Theta(\sqrt{X/f(\sqrt{X})})$ gives $X \leq K^2/2 + X\varepsilon_K + o(X)$ where $\varepsilon_K = \sum_{k > K} 1/(kf(k)) \rightarrow 0$ by the convergence hypothesis. Since $K^2 = o(X)$ (from $f(\sqrt{X}) \rightarrow \infty$) and $\varepsilon_K \rightarrow 0$, we obtain $X \leq 3X/4$ for X large — a contradiction. (Full proof in `lower_bound_io.tex` of the companion repository.) $\square$

Remark. *The partition-identity argument is sharp at the convergence boundary $\sum 1/(nf(n)) < \infty$: at $f(n) = \log n$ (where the sum diverges), the tail ε_K no longer goes to zero with K and the contradiction collapses. Pushing past requires either a sharper density estimate in the anchor window or a non-partition counting identity. A computational probe (`fourier_probes.py`) on greedy data verifies the Fejér PDS identity $\int_0^1 K_T(\xi) |\widehat{1_A}(\xi)|^2 d\xi = |A| + T - 1$ exactly, and shows the anchor window count is uniformly $\approx 0.42 \times |A|/\sqrt{\max A}$ — a constant multiple of greedy’s overall Sidon density, with no exploitable PDS-specific slack.*

Remark (Typographical note). *At the time of writing, erdosproblems.com/1194 states the conclusion of Theorem 3 under the condition $\sum 1/(nf(n))$ diverges. This appears to be a copy-edit error: the proof requires convergence, and so does Bloom’s example $f(n) = (\log n)^2$ in the comment thread (post 5784).*

4 Exact small- N optima

4.1 The finite problem

For each $N \geq 1$, let

$$M^*(N) := \min\{\max(A) : A \subset \mathbb{N}, A \text{ Sidon}, \{1, M\} \subset A, [1, N] \subset A - A\}.$$

This is the minimum value of $\max(A)$ over all finite Sidon sets that realise every difference in $[1, N]$ at least once. Trivially $M^*(N) \geq N + 1$ and $M^*(N) \geq 1 + \binom{K}{2}$ where $K = \lceil (1 + \sqrt{1 + 8N})/2 \rceil$ is the smallest K with $\binom{K}{2} \geq N$.

4.2 Exact values

We computed $M^*(N)$ via two engines: a Python bitmask depth-first search (`exact_search.py`) and a Google ortools CP-SAT model (`exact_search_cpsat.py`). For each N , both the feasibility at $M^*(N)$ and the infeasibility at $M^*(N) - 1$ are verified exhaustively. Lev’s greedy PDS (`explore.py`) gives an upper reference.

N	M^* exact	greedy max A	gd / M^*	$ A^* $	$ A_{\text{gr}} $
20	36	97	$2.7\times$	8	11
25	46	225	$4.9\times$	9	15
30	56	444	$7.9\times$	10	19
35	56	625	$11.2\times$	10	21
40	86	1175	$13.7\times$	12	27
45	86	1492	$17.4\times$	12	29
50	≥ 92	2518	$\geq 27.4\times$	—	35

Table 1: Finite PDS optima vs greedy. M^* verified to optimality for $N \leq 45$; $N = 50$ proves $M^* > 91$ (CP-SAT exhausts at $M = 91$ infeasible, $M = 92$ times out at 35 s/M).

Proposition 1 (Empirical scaling). *Across the data of Table 1, $M^*(N)/N \in [1.60, 2.15]$, hovering close to the trivial floor $N + 1$.*

The empirical conclusion is $M^*(N) \approx 2N$. By Sidon density, $|A^*| \approx \sqrt{2N}$, and indeed Table 1 shows $|A^*|$ within 1 of $\lceil (1 + \sqrt{1 + 8N})/2 \rceil$.

Witness sets (sorted) for the first few N :

- $N = 20$: $\{1, 5, 6, 18, 20, 26, 29, 36\}$.
- $N = 30$: $\{1, 2, 7, 11, 24, 27, 35, 42, 54, 56\}$.
- $N = 40$: $\{1, 10, 11, 18, 31, 43, 46, 57, 62, 80, 84, 86\}$.

5 The extension obstruction

The finite optima of Section 4 are extremal: they pack $\binom{|A^*|}{2}$ distinct differences into roughly $[1, 2N]$, saturating Sidon density. We now show empirically that these extremal seeds are *terrible starting points* for extending to a larger PDS.

5.1 Extension cost

For a seed A_{seed} (a Sidon subset of $\mathbb{N}$), define

$$M_{\text{ext}}(A_{\text{seed}}, N) := \min\{\max(A) : A \supseteq A_{\text{seed}}, A \text{ Sidon}, [1, N] \subset A - A\}.$$

Each step is again solved to optimality via CP-SAT (`extension_cost.py`). Two trajectories were computed:

Log-log fits on the “hard” steps (those requiring new elements) give:

- Seed A_{20}^* : $M_{\text{ext}} \sim N^{3.80}$ overall, $N^{6.48}$ in the tail.
- Seed A_{30}^* : $M_{\text{ext}} \sim N^{4.27}$ overall, $N^{4.90}$ in the tail.

5.2 Interpretation

We summarise the chain:

$$M_{\text{unconstrained}}^*(N) \sim N \ll M_{\text{greedy}}(N) \sim N^3 \ll M_{\text{ext, seeded}}(N) \gtrsim N^4. \quad (1)$$

N	M_{ext} from A_{20}^*	M_{ext} from A_{30}^*	greedy	unconstrained M^*
20	36	—	97	36
30	144	56	444	56
40	344	168	1175	86
45	1054	254	1492	86
56	—	516	3358	—
60	—	1019	—	—
63	—	1590	—	—

Table 2: Extension cost from finite optima vs scratch greedy vs unconstrained M^* .

A finite extremal optimum $A_{N_0}^*$ packs $\binom{|A|}{2}$ distinct differences into a window of width $\sim 2N_0$, leaving very few free difference slots. Every subsequent element added must avoid a densely packed set of forbidden differences. Greedy avoids this trap by being deliberately conservative (every Strategy-2 element lives at $\geq \max(A) + n + 1$), keeping the difference budget loose at the cost of a suboptimal finite $\max(A)$.

5.3 A conjecture for the infinite problem

The chain (1) is consistent with the i.o. lower bound $a_n \gg n^{2-o(1)}$ of Theorem 2, and in fact strengthens it empirically: a *naive* gluing of finite optima produces a_n growth strictly worse than N^3 . Beating Lev’s greedy requires a construction that balances tightness at current N against future extensibility — a non-local property.

This motivates the following “for all n ” analog:

Conjecture 1 (Extension lower bound). *There exists $c \in (1, 3]$ such that for every infinite PDS A , $a_n \gg n^c$ for all sufficiently large n . The value of c is governed by the asymptotic exponent of $M_{\text{ext}}(A_0, N)$ as the seed A_0 ranges over Sidon sets with $\max(A_0) \rightarrow \infty$.*

Conjecture 1 is strictly stronger than the i.o. form of Theorems 2 and 3. Empirically the extension-cost trajectories suggest c is a moderate constant (somewhere between 2 and 3), but the data does not distinguish a specific value.

6 Constructive upper bound and the cubic ceiling

6.1 Lev’s greedy and its constant

Lev [3] proves that the greedy construction (always choose the smallest valid element to cover the next missing difference) produces an infinite PDS with $a_n \leq Cn^3$ for some $C > 0$. Empirically [5] we observe

$$\frac{\max_{n \leq N} a_n}{N^3} \rightarrow 0.0293 \pm 0.0006 \quad (N \in [1500, 1900]).$$

6.2 Dense-Sidon-seeded greedy

Replacing the empty initial A with a known dense Sidon set (Mian–Chowla, Erdős–Turán, or a Phase 4 optimum) and running the same Strategy 1 / Strategy 2 greedy as Lev’s gives a strictly better constant:

Three observations from Table 3:

seed	$ A_{\text{seed}} $	N	seeded max A	ratio $/N^3$
scratch greedy	0	500	3,080,237	0.02464
Erdős–Turán $p = 23$	23	500	544,923	0.00436
Mian–Chowla $n = 30$	30	500	368,611	0.00295
Mian–Chowla $n = 60$	60	500	286,281	0.00229
Mian–Chowla $n = 100$	100	500	423,111	0.00339
Phase 4 opt A_{30}^*	10	500	881,451	0.00705

Table 3: Dense-Sidon-seeded greedy: the constant in the N^3 scaling drops by up to $10.8\times$ when seeded with Mian–Chowla of size 60. The Phase 4 optimum is a *bad* seed.

1. Mian–Chowla seeding reduces the empirical constant from $\approx 0.025 N^3$ to $\approx 0.0023 N^3$.
2. The Phase 4 optimum A_{30}^* , despite being the densest possible Sidon set covering $[1, 30]$, is a *worse* seed than Mian–Chowla — giving ratio 0.00705 vs 0.00229. This is a direct empirical witness of the extension obstruction (Section 5): tight seeds are extension traps.
3. No seed produces sub-cubic asymptotic growth. The constant moves; the $\Theta(N^3)$ scaling does not. Achieving $a_n = o(n^3)$ (Cilleruelo–Nathanson Problem 1 of [4]) requires a construction that is fundamentally non-greedy.

6.3 Block-structured seeds: a systematic no-go

To test whether the cubic ceiling is an artifact of *single-block* seeding, we built two block-structured families (`block_construction.py`) and completed each with the same greedy continuation:

split_MC(k_1, k_2) Partition $\text{MC}_{1\dots k_1+k_2}$ (Mian–Chowla) into two disjoint chunks and translate the second by a gap exceeding $\max(\text{within-block differences}) + \max(B_1)$, so cross-block differences avoid within-block differences. Because Mian–Chowla is globally Sidon, the two chunks share no within-block differences, and the union is Sidon.

AP_MC(k, m) Split $\text{MC}_{1\dots km}$ into m chunks of size k placed along an arithmetic progression of offsets chosen to avoid all cross-chunk collisions.

strategy	$ A_{\text{seed}} $	seed max	max A at $N=500$	ratio $/N^3$
scratch greedy	0	—	3,080,237	0.02464
MC(60) (baseline)	60	7,547	286,281	0.00229
split_MC(30,30)	60	15,012	280,501	0.00224
split_MC(40,40)	80	31,232	331,746	0.00265
split_MC(60,30)	90	43,655	386,310	0.00309
AP_MC($k=20, m=3$)	60	31,887	383,640	0.00307
AP_MC($k=30, m=2$)	60	21,247	302,887	0.00242

Table 4: Block-structured seeds at $N = 500$. The best variant (split_MC(30,30), ratio 0.00224) is statistically indistinguishable from the unsplit MC(60) baseline (0.00229). No construction breaks the cubic ceiling; the constant floors near $0.002 N^3$.

The reason is now empirically clear: once the seed is fixed, the greedy *continuation* is forced into Strategy-2 placements whose per-call cost is $\Theta(N^2)$, independent of seed structure. This drives the cubic. Beating $o(n^3)$ therefore requires redesigning the continuation, not the seed — e.g. probabilistic placement with structured rejection sampling, or algebraic constructions (projective-plane difference sets over $\mathbb{F}_q$ lifted to $\mathbb{Z}$).

7 Open problems and the $c = 2$ barrier

7.1 What is sharp

Two boundaries appear sharp in our analysis:

- The partition argument is sharp at the convergence boundary $\sum 1/(nf(n)) < \infty$ (Remark 3.3). The computational probes find no slack in Step (II) beyond the Sidon-density factor.
- Greedy-style construction is sharp at $\Theta(N^3)$ across all seeds tested (Tables 3 and 4), including block-structured seeds. Seeding gives constant-factor improvements; the exponent is robust.

7.2 Density attempts that fail

A natural route past the partition argument is a sharper Sidon density bound specific to PDS: Lindström gives $k(M) \leq \sqrt{M} + O(M^{1/4})$ for any Sidon set, while Cilleruelo–Nathanson [4] construct a PDS with $\limsup k(x)/\sqrt{x} \geq 1/\sqrt{2}$. Showing $\limsup k(x)/\sqrt{x} \leq 1 - \delta$ for every PDS would be a first PDS-specific density theorem. We document three structural attempts (full write-up in `proofs/phase2a_density.tex`), all of which fail for an instructive common reason:

1. *Cauchy–Schwarz via the partition identity.* One hopes the perfectly-packed coverage $N(x) = \binom{k(x)}{2}$ forces structure; but a_n is non-monotone, so the *covered prefix* $T_M := \max\{T : [1, T] \subset A_M - A_M\}$ can be far smaller than $\binom{k(M)}{2}$ (greedy at scale 2×10^8 has $T_M = 1898$ against $\binom{2131}{2} \approx 2.27 \times 10^6$).
2. *Fejér-weighted Plancherel.* For PDS, $\int_0^1 K_T(\xi) |\widehat{1_{A_M}}(\xi)|^2 d\xi = k + T - 1$ for $T \leq T_M$ — an exact identity, verified numerically. But the only available upper bound on the integral is the pointwise $|\widehat{1}|^2 \leq k^2$, which returns the trivial Sidon constraint $T \leq k^2$.
3. *Higher L^p moments.* For $p \geq 4$, $\int |\widehat{1_{A_M}}|^{2p}$ counts difference-pattern solutions that for Sidon sets are determined by $|A|$ alone; the PDS property adds no constraint at the moment level.

The structural reason all three fail: the PDS-vs-Sidon distinction lives in *which* small differences appear (every $n \leq T_M$ exactly once), not in any aggregate counting quantity. Moment and Cauchy–Schwarz arguments see only aggregates. The partition identity of Lemma 2 is the unique elementary tool that uses “which differences appear,” and it caps at $c = 2$ (Remark 3.3).

Three sub-targets remain open and may be more tractable: (i) the exceptional set $\{x : k(x) > (1 - \delta)\sqrt{x}\}$ has density 0 in $\mathbb{N}$; (ii) close the gap between $1/\sqrt{2}$ (CiNa08) and 1 (Lindström) for $\limsup k(x)/\sqrt{x}$ over PDS; (iii) strengthen the i.o. form of Theorem 2 to a positive density of n .

7.3 The three open questions

1. **Better i.o. bound.** Can $a_n \gg n^c$ i.o. be shown for some explicit $c > 2$? Phase 2B suggests this requires restriction / decoupling estimates or a non-partition counting identity.
2. **Filleruelo–Nathanson Problem 1.** Does there exist a PDS with $a_n = o(n^3)$? Sections 6 and 6.3 establish empirically that the $\Theta(N^3)$ ceiling holds under *any* greedy continuation regardless of seed — single dense blocks, split blocks, and AP-of-blocks all floor at $\approx 0.002 N^3$. A sub-cubic PDS, if it exists, must place elements non-greedily, anticipating future coverage; probabilistic and algebraic (projective-plane) constructions remain untested.
3. **For-all- n lower bound.** Is Conjecture 1 true? Even an explicit $c > 1$ would be the first “for all large n ” improvement on the trivial $a_n \geq n + 1$.

7.4 Beyond partition: candidate routes

Honest assessment of where to look:

- For (1), Section 7.2 rules out Cauchy–Schwarz, Fejér-weighted Plancherel, and L^p -moment arguments: each collapses to the generic Sidon bound. What remains is genuinely non-elementary — restriction/decoupling estimates specific to PDS (Bourgain–Demeter-style decoupling has been used for Sidon sets; a PDS analog would be new), or a non-partition identity that sees *which* differences appear.
- For (2), block-structured constructions are now tested and ruled out (Section 6.3): concatenating Mian–Chowla chunks with collision-free offsets does not move the exponent. The remaining candidates are constructions that abandon greedy continuation altogether — probabilistic placement with structured rejection sampling, or lifts of finite projective-plane (Singer) difference sets.
- For (3), one would convert the empirical Conjecture 1 into a theorem by showing: any infinite PDS truncated to $[1, M]$ produces a Sidon set A_M whose “extension obstruction” is at least polynomial. The Phase 4’ data is suggestive but we do not see a path to proof.

Acknowledgements, attribution, and code

This note records work done in the repository https://github.com/memoatwit/erdos_1194. Specifically:

`explore.py`, `build_checkpoint.py` Lev’s greedy.

`exact_search.py`, `exact_search_cpsat.py` Phase 4 finite-optimum search.

`extension_cost.py` Phase 4’ extension cost.

`seeded_greedy.py` Phase 3’ dense-Sidon-seeded greedy.

`block_construction.py` Phase 6 block-structured seeds.

`fourier_probes.py` Phase 2B Fourier observables.

`verify_and_analyze.py` Phase 0 / 1 verification + stratification.

The arguments of Theorems 2 and 3 are due to GPT-5.4 Pro and T. F. Bloom (April 2026 comment thread on erdosproblems.com/1194); we record full proofs with permission of the public thread. The remainder is original to this work.

Generative-AI disclosure. Large language models (Anthropic Claude, accessed via the Claude Code agent interface, and OpenAI models, 2026) were used for drafting assistance, code scaffolding, and consistency checking under the author’s direction. All mathematical claims, experiment designs, and conclusions were reviewed and verified by the author.

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